

Homework ORCA

- *This Homework Assignment consists of six (6) exercises: 4 computational exercises and 2 theoretical questions.*
- *Please prepare a report containing the answers, solutions and explanations for these exercises.*
- *Software to use for this report: **ORCA 5 or ORCA 6** and any visualisation software of your choice: **Avogadro, Chemcraft**, etc.*
- *Deadline to submit the report: **Sunday, 27 September 2026, 23h59m.***
- *The report must consist of **a single PDF file** for each student.*
- *A maximum of 5 (five) upload attempts are allowed before the deadline. The latest uploaded file will be used for verification.*
- *This Homework Assignment weighs for **15%** of the final grade for the CM5235 module.*

Instructions:

- *Be concise in your answers. Write answers exactly to the point.*
- *Try to achieve the results individually. Copying someone else's answers, reports, or computational results does not help your own development and understanding.*
- *Plots must be created by Origin, Excel, Mathematica, Python, or equivalent—no hand-drawn plots are allowed.*

Full Name : _____

Student Number: _____

E-mail: _____

Date _____

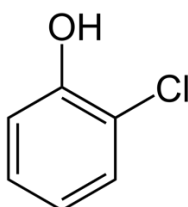
Signature _____

Download from CANVAS the files containing the XYZ atomic coordinates of the all the molecules used in this exercise set.

You may consult the website <https://www.faccts.de/docs/orca/6.0/manual/> for documentation, examples, related to ORCA calculations.

Exercise 1:

Study of the ground state of the 2-chlorophenol (ortho-chlorophenol molecule, C_6H_5Cl-OH) with different computational methods



Perform a DFT molecular structure relaxation of the **2-chlorophenol molecule**. In the ground electronic state. Use **def2-TZVP** basis set and **spin unrestricted** method + the following:

- **PBE** functional
- **B3LYP** functional
- **PBE** functional + solvent effects (water)
- **B3LYP** functional + solvent effects (water)

Search in the output the following quantities and show the comparison. Make a table showing these quantities for the two calculations.

- a) **Mulliken** and **Loewdin** charges on the Cl, O, C, H, atoms.
- b) Vibrational frequencies
- c) Electric dipole moment
- d) Entropy at room temperature
- e) Determine HOMO-LUMO gaps.

It is advisable to make a table comparing the above results.

Visualize the **normal mode number 12** and save it as a small video (or animated GIF, or equivalent).

In Avogadro, the function of saving the video is not working properly. For this case, you may use a "screen saver/recording" software. As an alternative, one can use other software, such as Chemcraft or VMD.

Plot HOMO and LUMO molecular orbitals for the **2-chlorophenol molecule** for one of the computational methods.

Plot the electronic density together with the electrostatic potential for the **2-chlorophenol molecule** using Avogadro or Chemcraft for one of the computational methods.

Exercise 2

Theoretical question #1

How are the solvent effects accounted for in the calculations done in **Exercise 1**? Write the details of the underlying computational method employed. Which other computational approaches for accounting for solvent effects can you use in ORCA? (see here <https://www.faccts.de/docs/orca/6.0/tutorials/prop/cpcm.html>)

Exercise 3:

Theoretical question #2

Describe what is the essence of the “Resolution of Identity” groups of methods (also known as “density fitting”. These methods allow a significant speedup of all ab initio calculations. How is this acceleration achieved? (Hint: read about this topic in the ORCA manual)

Exercise 4

Excited states and spectra

For the *optimized* molecular structures in Exercise 1, use the **time-dependent TD-DFT** to calculate the first 5 (five) excited states of the **2-chlorophenol** molecule.

Make a table to compare the four sets of energies of the excited states. How much do the solvent effects influence the results? Plot the UV spectra:

Read more here:

- 1) <https://www.faccts.de/docs/orca/6.0/tutorials/spec/UVVis.html>
- 2) <https://sites.google.com/site/orcainputlibrary/excited-state-calculations/tddft>

Exercise 5:

Calculation of magnetic exchange using BS-DFT

- Calculate the magnetic exchange interaction between Cu-Cu in the provided **Cu₂** compound. The compound has a total charge -2 . Each of the Cu atoms has an oxidation (+2, $3d^9$ configuration) and the spin ground state of each Cu is $S = 1/2$.
- Repeat the same calculation in Exercise 5, but include the **relativistic effects** at **DKH** and **ZORA** levels of theory. For this reason, use the relativistic basis sets DKH-def2-SVP and ZORA-def2-SVP, respectively.

Make a table comparing the obtained total energies and magnetic exchange values in all these three calculations. What can you learn from this table? How important are the relativistic effects on the computed value of magnetic exchange?

Read more here:

- 3) <https://sites.google.com/site/orcainputlibrary/dft-calculations/broken-symmetry-dft>
- 4) <https://sites.google.com/site/orcainputlibrary/relativistic-approximations>

Exercise 6:

Ab initio and semi-empirical molecular dynamics simulations

Run a short **molecular dynamics** simulation at

- a) DFT, PBE, def2-SVP, and
- b) semi-empirical (XTB) level of theories

of a **cluster of water** molecules in a sphere of size 50 Angstroms at **400 K**, using the Nosé–Hoover chain thermostat. Use the ORCA “%md” module to record one trajectory of 200 steps (for a) and 5000 steps (for b).

At this temperature, the original cluster, which is obtained from an ice crystal, starts to melt, the periodicity is broken, and its arrangement becomes disordered.

Open one of these trajectories using Avogadro and save it as a movie (or animated GIF).

Please use the small water cluster uploaded on the Canvas for DFT dynamics. For semiempirical XTB dynamics, you can use the large water cluster. What can you learn from the performed simulations? Which information can be further retrieved from the recorded trajectories?

You can read more and learn about ORCA MD calculations from:

- 1) <https://brehm-research.de/orcamd.php>
- 2) <https://sites.google.com/site/orcainputlibrary/molecular-dynamics>